BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

LESSON 069 · Lesson 069 · Data Mining and Visualisation of Patterns

Name: _____ Class: _____ Date: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

Do Now · Recall previous learning

Q1. What is a backtracking branch?

Q2. What is an outlier?

Q3. What is the purpose of a scatter graph?

Q4. Array [2,3,4,5]: double elements at indexes 1 and 3. Give updated array and sum.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

I Do · Read new knowledge

Key vocabulary

data mining — Finding patterns in datasets.

scatter plot — Pairs of numeric variables plotted as points.

outlier — Unusually distant observation.

correlation — Association, not proof of causation.

New learning

Define and predict — Data mining seeks patterns or associations across sets of data; define the question, prepare records and examine the results.

Explain — Visualisation makes patterns and anomalies visible. Scatter charts suit pairs of numeric data; bar charts compare categories.

Evaluate — A correlation does not establish cause. Sampling, confounders, missing values and measurement choices may mislead interpretation.

Worked example

For points (1,2),(2,4),(3,6), state trend and one limitation.

1. Follow the reasoning — Separate observation from causal explanation.

We Do · Practise with guidance

For scores [40,42,46,96], calculate mean and median; explain influence of outlier.

```
Mean (40+42+46+96)/4=56; median (42+46)/2=44.
```

Follow the reasoning

The value96 pulls mean away from three clustered lower scores.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

Question 1

For points (1,2),(2,4),(3,6), choose a suitable chart and describe association. [4]

Question 2

For [40,42,46,96] calculate mean and median with working. [6]

Question 3

Evaluate mining a school's student-usage logs to predict attainment. Address fairness, data quality and privacy. [6]

Exit Ticket · Answer before checking

A rising scatter trend proves that extra practice alone caused higher marks.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

Page 1 onwards · Do Now

Q1. One choice explored by search.

Q2. Value unusually different from others.

Q3. Displays association between two numerical variables.

Q4. [2,6,4,10]; total22.

You Do pages · Independent questions

Q1. Scatter plot [2]; increasing positive/linear association [2].

Q2. Sum224 [1], mean56 [2]; middle42,46 [1], median44 [2].

Q3. Unreliable or incomplete records [2], proxies/bias [1], privacy and minimisation [1], human review [1], correlation not proof of causation [1].

Exit Ticket · Model answer

False. Other factors may explain the relationship.

Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

1. Copy the whole original question:

2. Write a complete new answer in GREEN PEN. Show working if needed: